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IRAW: Novel invisible and robust adversarial watermark perturbations for digital image protection

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ABSTRACT

Invisible watermarking has long been a cornerstone for copyright protection due to its imperceptibility and forensic traceability. However, traditional watermarking remains a passive defense that fails to preclude unauthorized AI-driven analysis, such as the automated categorization and indexing of private media by illicit scrapers. To address this, we propose invisible and robust adversarial watermarking (IRAW), a unified framework that transitions image protection from passive traceability to proactive defense. By leveraging the discrete cosine transform (DCT) domain, IRAW ensures native compatibility with the JPEG compression protocols prevalent on mainstream social media platforms. The framework treats watermark embedding and adversarial perturbation generation as a synergistic optimization task. Specifically, an evolutionary optimization mechanism based on the integration of basin hopping (BH) with crossover and adaptive mutation operators is employed to navigate the search space and identify optimal perturbation patterns. To further enhance security, we implement a cross-domain defense architecture featuring dual spatial-domain encryption for host images and watermarks, combined with frequency-domain coordinate obfuscation. This mechanism not only bolsters cryptographic strength but also increases the structural complexity of perturbations, assisting the BH algorithm in escaping local optima to improve perturbation success rates. Experimental results demonstrate that IRAW generates high-fidelity adversarial examples with exceptional robustness against common image operations, such as JPEG compression and noise, while enabling reliable watermark recovery for forensic provenance. These findings establish IRAW as an effective and industrially compatible mechanism for modern digital image protection.

1. Introduction

With the rapid proliferation of social media platforms such as Weibo, Instagram, and Pinterest, a staggering volume of user-generated content (UGC) is shared globally every day. These digital images represent the intellectual and private property of creators, necessitating robust protection mechanisms against unauthorized exploitation (Khan, Tariq, Khalid, Mumtaz & Abdullah, 2023; Shaliyar & Mustafa, 2024). Historically, invisible watermarking has been a cornerstone of digital image protection. Notably, the discrete cosine transform (DCT) domain is widely recognized as the industrial benchmark for multimedia security (Ahmed, 1991) because it serves as the foundational transform for the JPEG compression protocol. Since nearly all mainstream UGC platforms utilize JPEG as their predominant format for image storage and

transmission, DCT-based techniques offer inherent advantages in terms of imperceptibility and forensic capabilities within these ecosystems. However, traditional watermarking remains a passive defense; while it facilitates ownership identification, it cannot prevent unauthorized AI-driven analysis. On modern digital platforms, illicit scrapers can exploit standard deep neural network (DNN) architectures to automatically categorize, tag, and index private media content, leading to large-scale unauthorized data exploitation and privacy breaches.

Recent literature (Carlini & Wagner, 2017; Goodfellow, Shlens & Szegedy, 2014; Moosavi-Dezfooli et al., 2017; Moosavideezfooli, Fawzi & Frossard, 2016; Papernot, McDaniel, Wu, Jha & Swami, 2016; Su, Vargas & Sakurai, 2019) has revealed that DNNs are highly vulnerable to adversarial perturbations. By introducing meticulously designed perturbations that are virtually imperceptible to the human eye, these

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models can be misled into making incorrect predictions. This vulnerability offers a unique opportunity for proactive privacy protection. Specifically, adversarial perturbations can be leveraged to neutralize unauthorized AI analysis by thwarting the ability of illicit scrapers to accurately categorize or index private images. Consequently, there is a significant potential to synthesize adversarial functionality with watermark embedding into a unified framework. This approach facilitates a transition from passive protection to a comprehensive defense strategy that couples proactive analysis-disabling with reactive traceability.

Early attempts by Jia et al. (Jia, Wei, Cao, & Han, 2020) utilized visible watermarks to generate adversarial perturbations; however, this approach inevitably introduces significant visual artifacts (see Fig. 1(a)). To enhance imperceptibility, subsequent research shifted toward invisible adversarial signals. For instance, schemes such as BHI (Liang, Feng, Chen, & Liu, 2023) and ISWP (Liang, Liu, Gao, Zhang, & Liu, 2025) leveraged least significant bit (LSB) embedding and the discrete wavelet transform (DWT) domain, respectively. Nevertheless, spatial-domain methods like BHI typically exhibit lower resilience than transform-domain techniques when subjected to common image processing attacks. Furthermore, while DWT-based schemes like ISWP offer improved stability, they face practical limitations regarding compatibility with mainstream social media platforms. Since platforms like Weibo and Instagram predominantly utilize JPEG—which is natively anchored in the DCT domain—media content is frequently subjected to mandatory re-encoding and lossy compression. Consequently, selecting an embedding domain that aligns with these ubiquitous industrial protocols is essential for ensuring that perturbation signals remain robust and effective in real-world web environments.

To address these challenges, we propose a novel invisible and robust adversarial watermarking (IRAW) method that implements a dual-functional mechanism within a unified optimization framework. In contrast to prior works, IRAW justifies the selection of the DCT domain from the outset to ensure seamless compatibility with the standard compression protocols of mainstream social media. In our scheme, the invisible watermark is designed to function as the adversarial perturbation, enabling a synchronized process that proactively disables unauthorized AI indexing while maintaining permanent copyright evidence for forensic provenance. Unlike linear frameworks that simply superimpose signals, IRAW treats watermark embedding and the generation of adversarial perturbations as a unified global optimization task. The IRAW framework integrates the basin hopping (BH) algorithm (Wales, & Doye, 1997) within an evolutionary optimization loop encompassing crossover, adaptive mutation, and selection (Goldberg, 1989; Storn, & Price, 1997). By iteratively refining the adversarial watermark through these combined mechanisms, IRAW achieves favorable robustness and visual fidelity. This ensures that the adversarial perturbations align with the practical requirements of social media scenarios and remain effective under various simulated distortions such

as JPEG compression and noise that characterize the typical re-encoding and transmission processes of digital platforms.

The main contributions of this study are summarized as follows:

- (1) Proactive image protection framework: We propose IRAW, a protection strategy that extends digital image security from passive traceability to proactive defense. The framework is designed to impede unauthorized DNN-based indexing of copyrighted images while maintaining reactive forensic capabilities through integrated invisible watermark signals.
- (2) Synergistic optimization via integrated mechanisms: The IRAW pipeline integrates basin hopping (BH) with evolutionary algorithm (EA) components and cross-domain encryption within a unified optimization task. The stochastic displacement mechanism of BH, combined with global evolutionary exploration, assists the algorithm in navigating the complex, non-convex search space. Additionally, cross-domain encryption introduces non-linear mappings that increase the complexity of the generated patterns, thereby enhancing the deceptiveness of the adversarial perturbations.
- (3) Robustness and dual-protection: Quantitative assessments verify the efficacy of this unified approach. Experimental results indicate that IRAW achieves high perturbation success rates while enabling reliable watermark recovery. Notably, the perturbation signals demonstrate sustained robustness against various simulated distortions such as JPEG compression and noise profiles that characterize the typical re-encoding processes of digital platforms.

The remainder of this paper is structured as follows. Section 2 provides a brief overview of related work on adversarial attacks and invisible watermarking. Section 3 provides the proposed IRAW method. Section 4 presents the experimental results and analysis. Section 5 concludes the paper and discusses future directions.

2. Relevant work

2.1. Copyright watermarking for multimedia security

The continuous expansion of user-generated content (UGC) within social media ecosystems (Kumar, Rahman, Namasudra, & Moparthy, 2024; Nawaz, Li, & Bhatti, et al., 2023) has made watermarking technologies indispensable for copyright protection and ownership verification (Dzhanashia & Evsutin, 2024; Yuan, Zhang, Wang, & Yin, 2024). These techniques embed secret information into host images in either visible or invisible forms (Melman & Evsutin, 2024). Invisible watermarking is particularly favored for digital image protection due to its transparency to the human visual system (Dzhanashia & Evsutin, 2024).

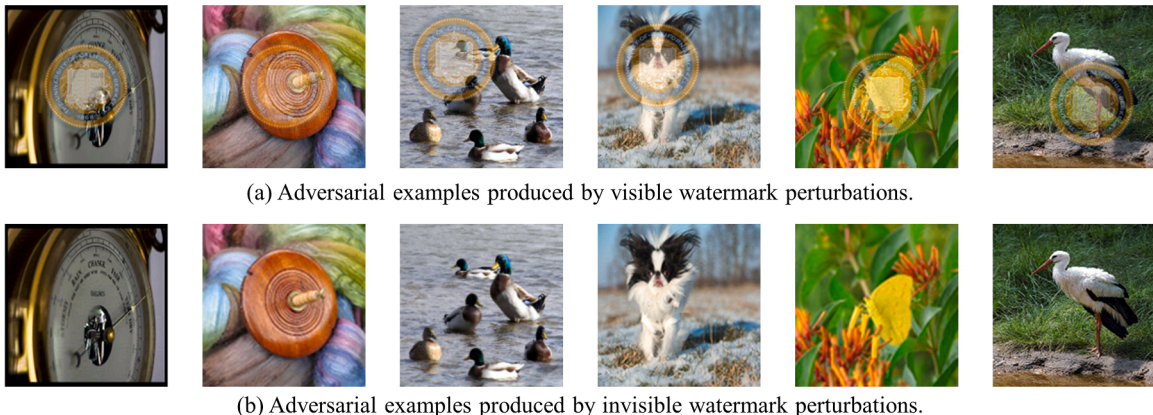


Fig. 1. Visual quality of adversarial examples with visible and invisible watermarks.

Traditionally, watermarks are embedded either in the spatial domain, such as the least significant bit (LSB) method (Goli & Naghsh, 2016), or in the transform domain using techniques like the discrete wavelet transform (DWT) (Alhaj, 2007) and discrete cosine transform (DCT) (Ahmed, 1991). While spatial methods are simple to implement, transform-domain watermarks typically exhibit superior resilience against common image processing attacks, such as noise addition and filtering. In the context of modern participatory media, a copyright watermark must not only be imperceptible but also robust enough to survive the lossy compression standards (e.g., JPEG) mandated by mainstream hosting platforms (Das, Shanbhogue, Chen, Hohman, Chen, Kounavis, & Chau, 2017). Recent studies (Dzhanashia & Evsutin, 2024; Melman & Evsutin, 2024; Yuan, Zhang, Wang, & Yin, 2024) have demonstrated a variety of data hiding techniques tailored to diverse application scenarios.

2.2. Adversarial perturbations

Deep neural networks (DNNs) have achieved state-of-the-art performance in image classification (He, Zhang, Ren, & Sun, 2016; Simonyan & Zisserman, 2014), yet they remain highly vulnerable to adversarial perturbations (Fawzi & Frossard, 2015; Goodfellow, Shlens, & Szegedy, 2014; Kanbak, Moosavi-Dezfooli, & Frossard, 2018). Since the introduction of the fast gradient sign method (FGSM) (Goodfellow, Shlens, & Szegedy, 2014), various perturbation strategies have been developed, including optimization-based methods like C&W (Carlini & Wagner, 2017) and iterative approaches like DeepFool (Moosavi-Dezfooli, Fawzi, & Frossard, 2016). Furthermore, specialized techniques have further exposed the fragility of these models, such as the one-pixel attack (Su, Vargas & Sakurai, 2019) that modifies as little as a single pixel via differential evolution, and the boundary attack (Wieland, Rauber, & Bethge, 2017) that navigates the decision space between adversarial and non-adversarial regions. While these methods effectively deceive classifiers, their perturbations typically manifest as semantically meaningless noise without practical significance. Moreover, because these "pure noise" patterns often rely on the precise manipulation of specific pixel values or delicate positions near decision boundaries, their robustness is often limited. Such adversarial signals frequently fail to survive standard image transformations or lossy compression attacks like JPEG (Das, Shanbhogue, Chen, Hohman, Chen, Kounavis, & Chau, 2017). This lack of robustness and semantic meaning limits their effectiveness in real-world scenarios, particularly on UGC platforms where media is frequently re-encoded.

2.3. Integration of copyright watermarks and adversarial perturbations

To bridge the technical gap between adversarial perturbations and digital content protection, researchers have begun exploring unified frameworks that synergistically integrate watermark embedding with adversarial example generation. This integration aims to generate adversarial examples that carry semantic meaning for copyright verification while maintaining the potency of adversarial functionality. Jia et al. (Jia, Wei, Cao, & Han, 2020) first utilized visible watermarks to generate adversarial perturbations, although noticeable visual artifacts remained. To improve imperceptibility, the BHI method (Liang, Feng, Chen, & Liu, 2023) employed invisible watermarking via the spatial-domain LSB technique, but its reliance on the spatial domain renders the adversarial signals vulnerable to common image processing attacks. Subsequently, the ISWP method (Liang, Liu, Gao, Zhang, & Liu, 2025) introduced DWT-domain watermarking to enhance robustness. While DWT-based schemes offer theoretical advantages in data hiding, they may face practical limitations regarding compatibility with mainstream social media platforms, which predominantly utilize JPEG compression as their standard for image storage and transmission.

To align with common image processing standards, our proposed IRAW method leverages the inherent properties of the frequency domain

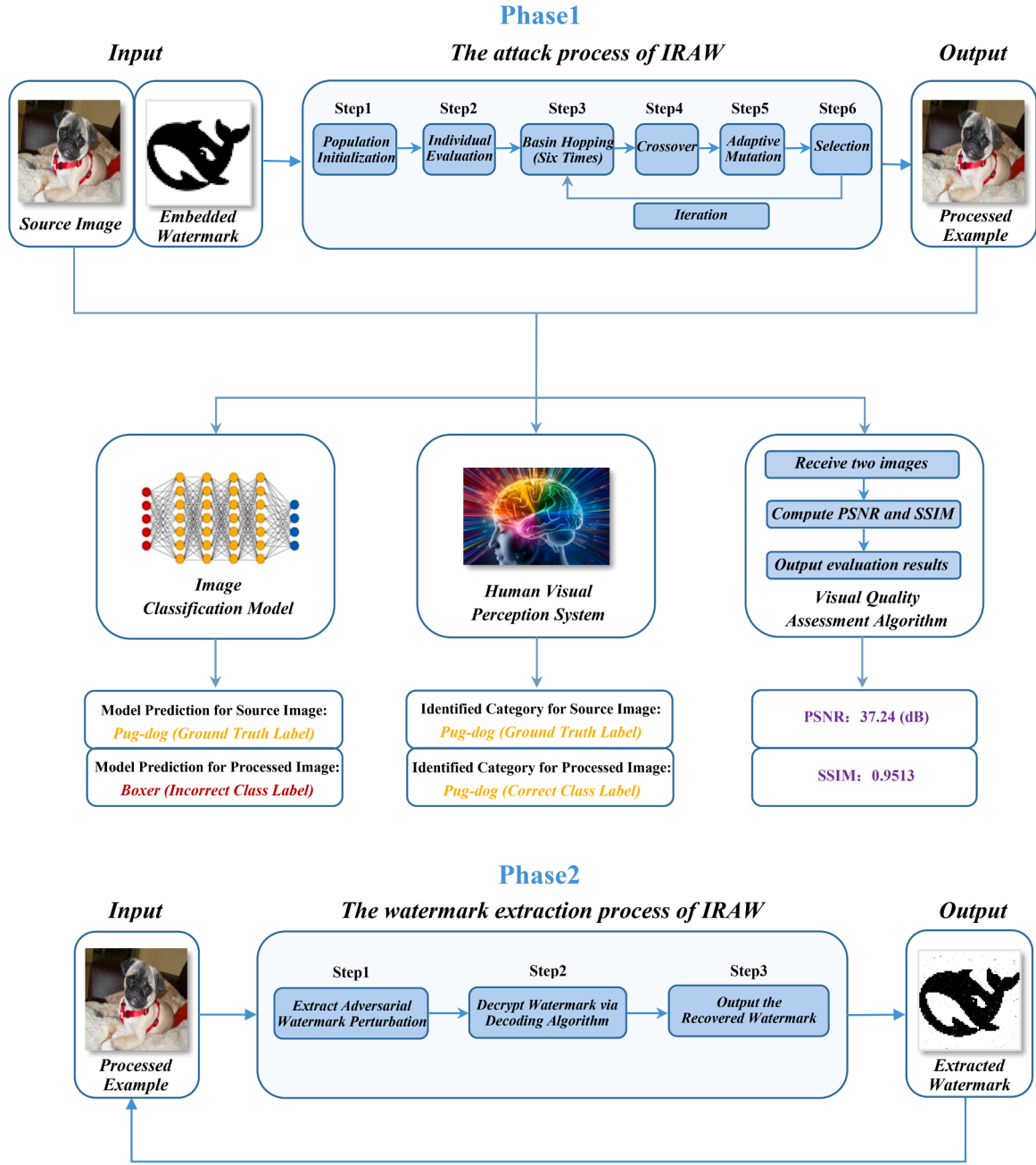
(Ahmed, 1991), thereby ensuring that both adversarial perturbations and watermark embedding exhibit strong resilience against simulated distortions. By treating these two objectives as a unified framework, the invisible watermark serves as the adversarial perturbation to proactively disable unauthorized AI analysis, such as illicit scrapers using AI for automatic categorization, while simultaneously preserving reactive traceability for post-event provenance. To implement this, we develop an optimization pipeline that integrates core evolutionary components, including population initialization, crossover, mutation, and selection (Goldberg, 1989; Storn & Price, 1997), with the basin hopping (BH) algorithm (Wales & Doye, 1997). This organic integration leverages stochastic mechanisms to explore a broader search space, which assists the optimization process in identifying superior global solutions within the discrete DCT domain. Furthermore, we introduce a cross-domain defense architecture that applies double encryption to the host image and watermark in the spatial domain, combined with frequency-domain coordinate obfuscation. The non-linear mappings and stochasticity introduced by this encryption process alter the statistical distribution of the signals, which not only helps the optimization navigate toward more effective global optima but also significantly enhances the perturbation success rate by increasing the complexity of the generated patterns.

3. Proposed method

Driven by the need for proactive privacy protection and industrial compatibility, this study proposes the invisible and robust adversarial watermarking (IRAW) method. By implementing the defense in the discrete cosine transform (DCT) domain, IRAW ensures native alignment with the foundational transform of JPEG standards, providing a compatible framework for image protection on mainstream user-generated content (UGC) platforms. The IRAW method integrates the basin hopping (BH) (Wales & Doye, 1997) algorithm within an evolutionary optimization loop (Goldberg, 1989; Storn & Price, 1997) (Fig. 2), utilizing stochastic mechanisms and global operators to identify optimal solutions in the search space. Furthermore, the framework adopts a cross-domain defense architecture that implements a double encryption mechanism for both watermark and host image data in the spatial domain, combined with frequency-domain coordinate obfuscation. While primarily designed to enhance system security, this cross-domain operation significantly increases the structural complexity of the generated adversarial perturbations. This heightened complexity provides the necessary stochasticity to assist the BH algorithm in escaping local optima more effectively during the optimization process. Consequently, the optimized distribution of the perturbation signals further misleads deep neural network (DNN) classifiers, thereby enhancing the perturbation success rate without compromising visual quality.

As illustrated in Fig. 2, the IRAW pipeline utilizes a source image and a watermark as inputs to produce an adversarial example with high visual fidelity. In Phase 1, the source image is accurately identified as "Pug-dog", whereas the processed example successfully deceives the classifier into outputting the incorrect label "Boxer". Both the source and processed images are visually indistinguishable as "Pug-dog" to human visual recognition systems. Quantitatively, the processed image achieves a peak signal-to-noise ratio (PSNR) of 37.24 dB and a structural similarity index measure (SSIM) of 0.9513, demonstrating superior visual fidelity. In Phase 2, the extraction process demonstrates that the embedded watermark can be perfectly recovered from the generated example. These results verify that IRAW fulfills its dual-protection design: the adversarial functionality provides a proactive defense against unauthorized AI-driven indexing, while the watermark ensures reliable copyright evidence.

In the following subsections, we present a comprehensive examination of IRAW through three key aspects: problem formulation, the proposed IRAW method, and the procedures for watermark embedding and extraction.



Constructing a Robust Defense Mechanism for Protecting Private Images

Fig. 2. Schematic diagram of the proposed IRAW framework.

3.1. Problem formulation

The issue of producing an adversarial example containing a watermark perturbation can be reformulated as a task of minimizing the classifier confidence that the generated adversarial example belongs to class t . Thus, our objective function is defined as follows:

$$\begin{aligned} & \min_{ft}(g(H, W, s, x, y, u, v, Key1, Key2, Key3, Key4, Ch)) \\ & \text{subject to } s \in [0, 15], x \in [0, W_H - w_W], y \in [0, H_H - h_W], u \in [2, 5], v \in [2, 5] \end{aligned} \quad (1)$$

In Eq. (1), H and W denote the host and watermark image respectively, with a scaling parameter s (ranging from 0 to 15) determined through extensive experimental validation. In this experiment, we employ 8×8 blocks for the discrete cosine transform (DCT). The

watermark (W) is embedded into the mid-frequency components of the host image (H) by modifying coefficients with frequency indices $u \in [2, 5]$ and $v \in [2, 5]$. Parameters x and y determine the embedding position of the resized watermark within the spatial domain of the resized host image. Their valid ranges are constrained by the dimensions of the resized host image (W_H, H_H) and the resized watermark (w_W, h_W), specifically $x \in [0, W_H - w_W]$ and $y \in [0, H_H - h_W]$, ensuring the watermark is fully embedded within the host image without exceeding its boundaries. The proposed method designs four dedicated encryption keys: Key_1 and Key_2 are dedicated to the dual encryption of the host image (H) and watermark (W); Key_3 and Key_4 are used for the obfuscation of frequency coordinates (u, v). This hierarchical key architecture ensures security across both spatial and frequency domains. The embedding channel selector Ch is encoded as a 3-bit binary value, where

Algorithm 1

IRAW: Invisible and robust adversarial watermarking.

Require: Host image: H ; Watermark image: W ; Target model: f ; Adversarial example: G ; Population size: Pop ; Dimension: 10; Max iterations: I .
Ensure: Adversarial example G ; Extracted watermark W_{ext} ; Global best solution X^* .

```

1. Procedure IRAW
2.   Define population  $P = \{X_1, \dots, X_{Pop}\}$ ;
3.   Define solution structure  $\mathbf{X} = \{s, x, y, u, v, Key_1, Key_2, Key_3, Key_4, Ch\}$ ;
4.    $h \leftarrow 0$ ; // Generation counter
5.   for  $i = 1$  to  $Pop$  do
6.     for  $j = 1$  to 10 do
7.        $X_{i,0,j} = X_{min,j} + \text{rand}(0, 1) \cdot (X_{max,j} - X_{min,j})$  // Population Initialization
8.     end for
9.   end for
10.   $X^* \leftarrow X_{1,0}$ ; // Initialization  $X^*$  after population generated
11.  while  $h < I$  do
12.    for  $i = 1$  to  $Pop$  do // Iterate through each individual  $X_{i,h} \in P_h$ 
13.       $\text{Fit-Val}(X_{i,h}) \leftarrow 1 - f_i(X_{i,h})$ ; // Current fitness
14.       $V_{i,h} \leftarrow \text{BasinHopping}(X_{i,h})$ ; // Local optimization
15.       $U_{i,h} \leftarrow \text{Crossover}(V_{i,h}, X_{i,h})$ ; // Crossover,  $nq\text{Uniform}\{1, \dots, Pop\} \setminus \{i\}$ 
16.       $M_{i,h} \leftarrow \text{Mutate}(U_{i,h})$ ; // Mutation
17.       $\text{Fit-Val}(M_{i,h}) \leftarrow 1 - f_i(M_{i,h})$ ; // Mutant fitness
18.      if  $\text{Fit-Val}(M_{i,h}) \geq \text{Fit-Val}(X_{i,h})$  then
19.         $X_{i,h} \leftarrow M_{i,h}$ ; // Update current individual
20.        if  $\text{Fit-Val}(X_{i,h}) > \text{Fit-Val}(X^*)$  then
21.           $X^* \leftarrow X_{i,h}$ ; // Select as the global best solution
22.        end if
23.      end if
24.    end for
25.     $h \leftarrow h + 1$ ;
26.  end while
27.   $G \leftarrow g(H, W, X^*)$ ; // Generate adversarial example
28.   $W_{ext} \leftarrow g'(G, X^*)$ ; // Watermark extraction
29.  return  $G, W_{ext}$ ;
30. End Procedure

```

001 corresponds to either the R channel in RGB space or Y channel in YUV space, 111 indicates all channels in either RGB or YUV space, and other combinations (100, 010, 110, 101, 011) enable flexible single-channel or multi-channel selection while ensuring that at least one channel of the host image (H) carries the watermark (W). Furthermore, we specify that 000 also denotes embedding across all channels in either RGB or YUV space. In practical implementation, the values of the three channels can either vary or remain constant. Specifically, the values of the three channels are kept constant when computational power and time costs are taken into consideration. In our study, $g(H, W, s, x, y, u, v, Key_1, Key_2, Key_3, Key_4, Ch)$ is abbreviated as $g(\cdot)$ that represents the generated adversarial example, and $f_i(g(\cdot))$ denotes the probability of predicting the adversarial example $g(\cdot)$ as label t . As a result, the primary objective of the adversarial perturbation is to significantly reduce the value of $f_i(g(\cdot))$.

3.2. Proposed IRAW method

The proposed IRAW method is a heuristic stochastic search algorithm that generates adversarial perturbations by locating the global minimum of a multivariate function, implemented through the six-step process illustrated in Fig. 2. The IRAW pipeline is an innovative synergistic composition designed to navigate the complex, non-convex search space of the DCT domain. While operators such as population initialization, crossover, and mutation are fundamental to evolutionary algorithms (EA) (Goldberg, 1989) and differential evolution (DE) (Storn & Price, 1997), the core contribution of IRAW lies in the purposeful fusion of these mechanisms with basin hopping (BH) (Wales & Doye, 1997). Although BH is inherently an iterative local search mechanism characterized by stochastic "jumping", its integration within the evolutionary cycle is what enables a comprehensive global search. By integrating BH within the population-based pipeline, the algorithm can effectively escape local optima in the frequency domain, leveraging BH's fine-grained exploitation alongside EA's global exploration to ensure an optimal trade-off between watermark robustness and perturbation

success. The complete six-step workflow of the proposed IRAW methodology is formally specified as follows:

Step1: Population Initialization

We randomly initialize a population of over 100 individuals. The i th individual in the h -th generation population is denoted as $X_{i,h}$ and the j -th gene of the individual $X_{i,h}$ is represented as $X_{i,h,j}$. The corresponding formulation is expressed as follows:

$$X_{i,0,j} = X_{min,j} + \text{rand}(0, 1) \cdot (X_{max,j} - X_{min,j}), i \in (100, 300), j \in [0, 10] \quad (2)$$

In Eq. (2), the j -th gene of the i th individual in the initial population is represented as $X_{i,0,j}$, while the minimum and maximum values of the j -th gene are denoted as $X_{min,j}$ and $X_{max,j}$ respectively. The individual $X_{i,h}$ corresponds to a potential solution in the search space, characterized by ten critical parameters: the scale factor (s), two frequency indices (u and v), position parameters (x and y), four encryption keys (Key_1, Key_2, Key_3 and Key_4), and the selected RGB or YUV channel combination (Ch) for embedding the watermark (W) into the host image (H). To strike a balance between computational efficiency and the frequency-domain characteristics of DCT, the value of Ch is set to 111 to enforce the watermark (W) embedding across all three channels in the YUV (or RGB) color space of the host image (H). As shown in Algorithm 1, X^* is initially set to $X_{1,0}$ after the population is generated, providing an initial baseline for tracking the global optimum.

Step2: Individual Fitness Evaluation

$$\text{Fit} - \text{Val}(X_{i,h}) = 1 - f_t(X_{i,h}) \quad (3)$$

The population fitness evaluation algorithm plays a pivotal role in our framework, facilitating the effective selection of suitable candidates for successive generations through computationally efficient assessments. As specified in Eq. (1), we define $f_t(H)$ as the probability that the host image (H) is classified as label t by the target model $f(\cdot)$. Accordingly, Eq. (3) defines $f_t(X_{i,h})$ as the classifier's confidence score for adversarial examples generated from solution $X_{i,h}$. In our experiments, individual fitness within the population is evaluated quantitatively using Eq. (3). The maximization of this fitness value (Fit-Val) indicates superior environmental adaptation capability. Based on this criterion, we implement an elitist selection strategy that propagates the top 100 highest-fitness individuals to the subsequent generation.

Step3: Basin Hopping

The basin hopping (BH) (Wales & Doye, 1997) stochastic optimization algorithm, which finds local minima via fitness function evaluation, evolves the original individual $X_{i,h}$ into an improved candidate $V_{i,h}$. Specifically, we employ BH to perform a synchronous search across 100 independent local regions, each corresponding to one of the 100 distinct individuals. The search center (given individual) explores the 10-dimensional space per region, with direction governed by an 10-dimensional Gaussian vector and step size controlled by a fixed stride length (FSL). This process is mathematically formulated as follows:

$$V_{i,h} = X_{i,h} + \text{FSL} * \text{vector}(\text{Gaussian}) \quad (4)$$

where $X_{i,h}$ and $V_{i,h}$ denote the original search center and the newly produced candidate in this search respectively; FSL and $\text{vector}(\text{Gaussian})$ determine the step size and the direction of m -dimensional space in the search operation respectively. The fitness values (Fit-Val) of both the generated candidate $V_{i,h}$ and the original individual $X_{i,h}$ are computed to evaluate their performance. The candidate $V_{i,h}$ replaces $X_{i,h}$ if and only if

it demonstrates superior fitness, as formulated below:

$$\text{BasinHopping}(X_{i,h}) = \begin{cases} V_{i,h} & \text{Fit} - \text{Val}(V_{i,h}) > \text{Fit} - \text{Val}(X_{i,h}) \\ X_{i,h} & \text{Fit} - \text{Val}(V_{i,h}) \leq \text{Fit} - \text{Val}(X_{i,h}) \end{cases} \quad (5)$$

Step4: Crossover

Two high-fitness parent individuals $V_{i,h}$ and $X_{n,h}$ undergo a crossover operation with a rate (CR) of 0.9. This process first involves gene recombination at each locus, where the j -th gene of the offspring $C_{i,h,j}$ is generated by recombining the j -th genes of the parents $V_{i,h,j}$ and $X_{n,h,j}$, following the rule:

$$C_{i,h,j} = \text{Recombine}(V_{i,h,j}, X_{n,h,j}) \\ \text{subject to } \text{rand}(0, 1) \leq \text{CR}, i \neq n \quad (6)$$

Upon completing recombination across all genes, a full offspring individual $C_{i,h}$ is formed. The crossover function in Eq. (7) then selects the fittest individual from the two parents ($V_{i,h}$ and $X_{n,h}$) and the offspring ($C_{i,h}$) by comparing their fitness values (Fit-Val):

$$U_{i,h} = \text{Crossover}(V_{i,h}, X_{n,h}) \\ = \begin{cases} V_{i,h} & \text{Fit-Val}(V_{i,h}) > \text{Fit-Val}(X_{n,h}), \text{Fit-Val}(V_{i,h}) > \text{Fit-Val}(C_{i,h}) \\ X_{n,h} & \text{Fit-Val}(X_{n,h}) > \text{Fit-Val}(V_{i,h}), \text{Fit-Val}(X_{n,h}) > \text{Fit-Val}(C_{i,h}) \\ C_{i,h} & \text{others} \end{cases} \quad (7)$$

Step5: Mutation

To generate a more suitable candidate ($M_{i,h}$), we employ mutation operation to change the genes of the individual ($U_{i,h}$), and the mutation process is implemented by the Eq. (8).

$$M_{i,h,j} = U_{i,h,j} + F * (X_{o,h,j} - X_{k,h,j}), F \in (0, 1), o \neq k \neq i \quad (8)$$

where $M_{i,h}$ is the individual mutated from $U_{i,h}$ that represents the solution achieved by the crossover operation. The j -th genes of $M_{i,h}$ and $U_{i,h}$ are denoted as $M_{i,h,j}$ and $U_{i,h,j}$, respectively. We randomly select two individuals ($X_{o,h}$ and $X_{k,h}$) to perform the mutation process. Their j -th genes are represented as $X_{o,h,j}$ and $X_{k,h,j}$, respectively. The genetic composition of $U_{i,h}$ is subject to random perturbations governed by the mutation operator F that is ranged from 0 to 1. As a result, a more suitable candidate ($M_{i,h}$) is generated.

$$M_{i,h} = \text{Mutate}(U_{i,h}) = \begin{cases} M_{i,h} & \text{Fit-Val}(M_{i,h}) \geq \text{Fit-Val}(U_{i,h}) \\ U_{i,h} & \text{others} \end{cases} \quad (9)$$

The $\text{Mutate}(\cdot)$ function, mathematically defined in Eq. (9), serves as an acceptance criterion for mutated individuals, determining their inclusion or exclusion in the subsequent population. During this process, a greedy selection is performed: the higher fitness individual between $M_{i,h}$ and $U_{i,h}$ is retained as the new $M_{i,h}$ for the subsequent step.

Step6: Selection

IRAW generates a new candidate solution $M_{i,h}$, which undergoes fitness evaluation against the current individual $X_{i,h}$ using the fitness metric. The replacement criterion is strictly as follows: $M_{i,h}$ is replaced by $X_{i,h}$ if the fitness value (Fit-Val) of $X_{i,h}$ is greater than or equal to that of $M_{i,h}$. If the updated $X_{i,h}$ achieves a higher fitness value than the global best solution X^* , it replaces X^* to become the new global optimal solution, ensuring the dynamic maintenance of the historical optimal state. This elitist selection mechanism ensures monotonic improvement in solution quality across iterations, thereby enhancing search efficiency.

$$X_{i,h+1} = \begin{cases} X_{i,h} & \text{Fit-Val}(X_{i,h}) \geq \text{Fit-Val}(M_{i,h}) \\ M_{i,h} & \text{others} \end{cases} \quad (10)$$

3.3. The processes of watermark embedding and extraction

3.3.1. Watermark embedding

The proposed method first processes the host image H and watermark image W through an adaptive resizing operation, where the scaling factor s is dynamically determined by the first gene parameter of the solution X^* . The specific embedding position (x, y) of the watermark in the spatial domain is dynamically determined by the optimization algorithm, and this design aims to enhance the effectiveness of adversarial perturbations while ensuring visual imperceptibility. The precise coordinate parameters are governed by the gene parameters in the solution space. Following this process, we apply distinct cryptographic parameters derived from the sixth and seventh components of the individual X^* to encrypt the host image H and watermark image W respectively. The IRAW method determines the appropriate combination of YUV (or RGB) channels, denoted as Ch , which corresponds to the final gene of the solution X^* . To achieve optimal balance between visual quality preservation and watermark robustness, our method strategically embeds watermarks exclusively within the mid-frequency DCT coefficients (where both u and v range from 2 to 5). This frequency area selection ensures minimal visual distortion and simultaneously maintains resistance against common image processing attacks. The selected coefficients are further obfuscated through encryption algorithms using dedicated keys (Key_3 and Key_4) prior to watermark embedding. The embedding process, formally defined in Eq. (11), generates the final adversarial example G through the function $g(\cdot)$, which seamlessly incorporates the encrypted watermark while preserving the host image's visual quality.

$$G = g(H, W, s, x, y, u, v, Key1, Key2, Key3, Key4, Ch) = g(H, W, X^*) \quad (11)$$

$$3 \leq L1 \leq L2 \leq 99, 2 \leq L3 \leq L4 \leq 5 \quad (12)$$

$$T1 = \log(L1 * L3 + L4), T2 = \log(L2 * L3 + L4) \quad (13)$$

$$Key1 = (T1 + L3) \bmod L1, Key2 = (T2 + L4) \bmod L2 \quad (14)$$

$$T3 = u * L3 + \log(L1 + L3), T4 = v * L4 + \log(L2 + L4) \quad (15)$$

$$Key3 = (T3 + L1) \bmod L3, Key4 = (T4 + L2) \bmod L4 \quad (16)$$

We present a detailed mathematical derivation of the cryptographic key generation algorithm supporting the IRAW watermarking framework. Eqs (12–16) define a specialized cryptographic key generation algorithm designed to produce four dedicated encryption keys, Key_1 to Key_4 , for the IRAW watermarking framework: the algorithm begins by initializing primary random seeds L_1 and L_2 within the integer range [3, 99], where L_1 is chosen to be no larger than L_2 to ensure a sufficiently large key space, while secondary constrained seeds L_3 and L_4 are selected from [2, 5] to provide a controlled yet non-trivial input to the key derivation process (Eq. (12)); intermediate values T_1 and T_2 are computed via logarithmic transformations of seed combinations (Eq. (13)) to introduce non-linearity and enhance key unpredictability, then combined with L_3 and L_4 respectively and reduced modulo L_1 and L_2 to generate Key_1 and Key_2 (Eq. (14)), which enable dual encryption of the host image and watermark; for DCT coefficient encryption, T_3 and T_4 are derived by combining the mid-frequency coordinate indices u and v with the seed values (Eq. (15)), and these intermediates are then used to generate Key_3 and Key_4 (Eq. (16)), which obfuscate the frequency coordinates to prevent unauthorized localization of the embedded watermark, with the integration of logarithmic operations and modular arithmetic ensuring that each key is pseudorandom and dependent on all input seeds, thereby increasing the complexity of key generation and strengthening the overall security of the system.

Fig. 2 demonstrates the application of our proposed IRAW framework in generating an adversarial example that maintains exceptional visual fidelity while misleading DNNs. The discrete cosine transform (DCT) serves as the fundamental transformation for watermark embedding, converting the spatial-domain host image H into its frequency-domain representation $C(u, v)$. The block dimensions M and N , configured as 8×8 pixels in this study, critically determine the frequency resolution and processing granularity. This is formulated as follows:

$$C(u, v) = \alpha(u) \alpha(v) \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} H(x, y) \cdot \cos\left(\frac{(2x+1)u\pi}{2M}\right) \cos\left(\frac{(2y+1)v\pi}{2N}\right)$$

$$\text{subject to } \alpha(u) = \begin{cases} \sqrt{\frac{1}{M}} & u=0 \\ \sqrt{\frac{2}{M}} & \text{others} \end{cases}, \alpha(v) = \begin{cases} \sqrt{\frac{1}{N}} & v=0 \\ \sqrt{\frac{2}{N}} & \text{others} \end{cases} \quad (17)$$

In this process, each pixel value $H(x, y)$ at spatial coordinates (x, y) , where x and y represent row and column indices respectively, is transformed into frequency coefficients $C(u, v)$. These coefficients are positioned at frequency coordinates (u, v) , with u and v indicating horizontal and vertical frequency components ranging from 0 to $M-1$ and $N-1$ respectively. The transformation employs normalization factors $\alpha(u)$ and $\alpha(v)$ to maintain energy conservation, scaling the DC component (where either u or v is equal to 0) for low-frequency representation while weighting the AC components for higher frequencies. The cosine basis functions perform the frequency decomposition, naturally clustering low-frequency components near the top-left (with small u, v values) and high-frequency components toward the bottom-right (with larger u, v values) of the coefficient matrix. In our watermarking process, the embedded data is primarily inserted into the mid-frequency region (where both u and v range from 2 to 5), as this area optimally balances robustness against compression with visual imperceptibility. Furthermore, we leverage the eighth and ninth genes from the solution $X_{i,h}$ to safeguard the frequency-domain coordinates (u and v).

When integrated with the preceding spatial-domain encryption mechanism, this dual-protection strategy establishes a comprehensive cross-domain defense mechanism. The combined approach simultaneously protects content integrity across both spatial and spectral dimensions, creating robust resistance against unauthorized access attempts. The design purpose of the host image encryption module is to integrate with watermark encryption and frequency-domain coordinate obfuscation, thereby constructing a cross-domain security system. Its core functions are as follows. First, it significantly elevates the security barrier for unauthorized attackers. Even if they possess the basic algorithm framework, they are unable to circumvent the multi-layered protection to acquire valid information, which effectively prevents the malicious replacement or forgery of watermarks. Second, it caters to the privacy protection requirements of social platforms by safeguarding sensitive content in images prior to watermark verification. For instance, encryption can prevent watermark tampering when professional creators release their works; for user-generated content (UGC) containing private information, encryption can avoid the extraction of privacy elements by third parties during web crawling. In the embedding phase, the host image undergoes 8×8 block DCT transformation according to Eq. (17) to obtain the frequency coefficient matrix $C(u, v)$. For coefficients in the mid-frequency region where both u and v range from 2 to 5, a coefficient modulation approach is adopted. Pre-encrypted binary watermark information is used to fine-tune coefficients at corresponding positions following preset mapping rules. Specifically, fusion is achieved via the function $g(\cdot)$ as shown in Eq. (11). The frequency coefficients are encrypted and protected using Key3 and Key4, and the adjustment amplitude is controlled to ensure visual imperceptibility and robustness against compression.

3.3.2. Watermark extraction

As the concluding step of IRAW, the ten essential genes ($s, x, y, u, v, Key_1, Key_2, Key_3, Key_4$, and Ch) of the solution $X_{i,h}$ are embedded within the last four pixels (P_1, P_2, P_3 and P_4) of the generated example G . Specifically: the scaling factor s and spatial coordinate parameters x, y are collectively encoded into the YUV (or RGB) channels of the first trailing pixel P_1 ; the frequency indices u, v , and encryption key Key_1 are embedded into the corresponding channels of the second trailing pixel P_2 ; the encryption keys Key_2, Key_3 , and Key_4 are stored in the YUV (or RGB) channels of the third trailing pixel P_3 ; and the YUV (or RGB) channel combination Ch is encoded into the designated channel of the fourth trailing pixel P_4 . To improve efficiency and reduce computational costs, Ch can typically be fixed at 111, i.e., all three RGB or YUV channels can be selected for watermark embedding. To enhance security against adaptive attacks, the extraction keys and parameters are pre-encrypted using a public key before embedding. Under this threat model, even if the specific embedding locations (e.g., the last four pixels) are identified, attackers cannot obtain the actual data without the authorized private key. This ensures the system's security is anchored in cryptographic strength rather than location secrecy.

$$W = g'(G, s, x, y, u, v, Key_1, Key_2, Key_3, Key_4, Ch) = g'(G, X^*) \quad (18)$$

Building upon this security framework, the extraction procedure initiates with a cryptographic verification phase: the four trailing pixels (P_1, P_2, P_3 , and P_4) of adversarial example G are first accessed to retrieve the encrypted metadata, which must be decrypted using the corresponding private key. Only upon successful decryption are these data entries processed through the decoding operations (see Eq. (18)) to reconstruct the optimal solution X^* , thereby enabling the precise recovery of all perturbation parameters ($s, x, y, u, v, Key_1, Key_2, Key_3, Key_4$, and Ch). During the subsequent watermark extraction phase, the watermarked adversarial example G undergoes a discrete cosine transform (DCT) to convert the spatial-domain pixels into frequency-domain coefficients. Leveraging the recovered u and v parameters, the mid-frequency region is accurately localized, after which the coefficients within this region are inverse-demodulated following the mapping rules adopted during watermark embedding. Subsequent decryption is conducted to recover the binary watermark information, which is further decoded to obtain the original watermark.

4. Experimental results and analysis

4.1. Experimental implementation details

The experimental evaluations in this study were conducted on the ImageNet dataset (Russakovsky et al., 2015) and CIFAR-10 dataset (Krizhevsky & Hinton, 2009), using the PyTorch framework with CUDA acceleration on an NVIDIA GeForce RTX 3090 GPU. For testing, 1000 images were randomly selected from each dataset. These images were evaluated on seven deep learning models: VGG19 (Simonyan & Zisserman, 2014), Inception V3 (Szegedy et al., 2016), ShuffleNet (Zhang, Zhou, Lin, & Sun, 2018), VGG16 (Simonyan, & Zisserman, 2014), ResNet50 (He, Zhang, Ren, & Sun, 2016), ResNet101 (He, Zhang, Ren, & Sun, 2016), and ResNet152 (He, Zhang, Ren, & Sun, 2016). These DNN models were employed as target models denoted as f in Eq. (1). For the generation of adversarial examples containing invisible and secure watermarks, two distinct watermark types were utilized: LOGO1 (a textual watermark) and LOGO2 (a geometrically-shaped watermark), as illustrated in Fig. 3. To ensure a comprehensive evaluation of the proposed IRAW method, comparative analyses were performed against prominent black-box adversarial perturbation methods, including single-pixel attack (Su, Vargas, & Sakurai, 2019), boundary attack (Wieland, Rauber, & Bethge, 2017), spatial attack (Engstrom, Tran, Tsipras, Schmidt, & Madry, 2017), pointwise attack (Schott, Rauber, Bethge, & Brendel, 2018), Jia et al.'s method (Jia, Wei, Cao, & Han, 2020), BHI

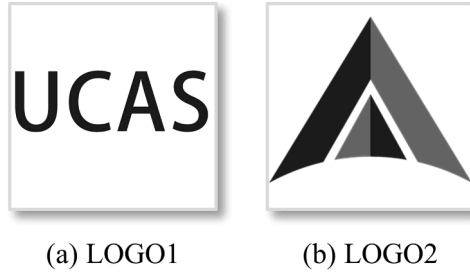


Fig. 3. Two categories of watermarks utilized in the experimental study.

Table 1

Success rates of IRAW using LOGO2 to perturb the VGG19 model under various hyper-parameter configurations of crossover probability (CR) and number of iterations (I).

	$I = 1$	$I = 2$	$I = 3$	$I = 4$	$I = 5$	$I = 6$	$I = 7$	$I = 8$	$I = 9$
CR = 0.5	54.3 %	58.7 %	63.0 %	67.4 %	71.7 %	76.1 %	80.4 %	78.3 %	80.4 %
CR = 0.6	56.5 %	60.9 %	65.2 %	71.7 %	78.3 %	76.1 %	78.3 %	80.4 %	80.4 %
CR = 0.7	60.9 %	65.2 %	69.6 %	69.6 %	73.9 %	78.3 %	76.1 %	78.3 %	80.4 %
CR = 0.8	58.7 %	60.9 %	67.4 %	69.6 %	73.9 %	76.1 %	78.3 %	80.4 %	76.1 %
CR = 0.9	63.0 %	71.7 %	71.7 %	73.9 %	78.3 %	80.4 %	80.4 %	76.1 %	78.3 %
CR = 1.0	65.2 %	69.6 %	73.9 %	71.7 %	76.1 %	78.3 %	80.4 %	80.4 %	78.3 %

(Liang, Feng, Chen, & Liu, 2023), and ISWP (Liang, Liu, Gao, Zhang, & Liu, 2025). An extensive series of preliminary experiments was conducted to optimize the hyper-parameters of the IRAW method, with a particular focus on the crossover probability (CR) and the number of iterations (I). As detailed in Table 1, these parameters were evaluated by using logo2 to generate adversarial examples for the VGG19 model. To achieve an optimal balance between temporal efficiency and adversarial performance, the hyper-parameters (CR and I) were ultimately set to 0.9 and 6, respectively. Under this configuration, the IRAW method attained its peak success rate of 80.4 %. These optimized parameter values were subsequently adopted for all following experimental evaluations across different models and watermark types.

4.2. Performance of the proposed IRAW method

4.2.1. Visual quality of adversarial examples

The proposed IRAW method effectively embeds invisible watermarks (LOGO1 and LOGO2) into original images, successfully generating corresponding adversarial examples while maintaining visual fidelity, as evidenced in Fig. 4 in which human observers cannot discern any noticeable differences between the original and generated examples. The comprehensive quantitative evaluations were conducted using two established metrics: peak signal-to-noise ratio (PSNR) and structural similarity (SSIM), both widely used metrics for image quality assessment, with higher values indicating better visual preservation. The quantitative results, presented at the bottom of Fig. 4, demonstrate superior visual fidelity. Among the six adversarial examples analyzed, the peak PSNR reaches 38.25 dB, while the overall average remains above 36.77 dB. Notably, even the case with the lowest PSNR (the fifth image) maintains a value of 35.97 dB, which is well beyond the threshold of visual imperceptibility. Similarly, SSIM evaluation reveals outstanding results, with the fifth image achieving an optimal score of 0.9859 compared to the minimum value of 0.9457 observed in the third image, and an impressive average SSIM surpassing 0.9696 across all test examples, confirming the adversarial examples generated by IRAW with invisible watermark perturbations have significant visual imperceptibility. Furthermore, in Section 4.3.2, a comprehensive quantitative comparison of visual quality is conducted across the entire dataset

against other state-of-the-art black-box perturbation methods. The results consistently demonstrate that our proposed method achieves favorable overall visual metrics.

4.2.2. Success rates of adversarial examples

Table 2 presents the success rates achieved by the proposed IRAW method using both textual (LOGO1) and geometric (LOGO2) watermarks across seven deep learning models. As illustrated in the table, the success rates exhibit a consistent upward trend as the process advances through the six progressive stages, demonstrating that iterative refinement effectively improves the overall success rate for both watermark types. Specifically, LOGO1 reaches an average success rate of 81.9 % at the final iteration ($I = 6$), showing notable effectiveness on architectures such as ShuffleNet (Zhang, Zhou, Lin, & Sun, 2018), ResNet50 (He, Zhang, Ren, & Sun, 2016), Inception_V3 (Szegedy et al., 2016), and VGG16 (Simonyan & Zisserman, 2014), where rates exceed 82 % and reach 94 % on ShuffleNet. While relatively lower success rates are exhibited on VGG19 (Simonyan & Zisserman, 2014), ResNet152 (He, Zhang, Ren, & Sun, 2016), and ResNet101 (He, Zhang, Ren, & Sun, 2016), these models still maintain an approximately 74 % success rate. Similar performance trends are evident for LOGO2 perturbations, with the average success rate progressively increasing from 69.3 % at $I = 3$ to 82.3 % by $I = 6$. Comparative analysis shows that both watermark types consistently yield over 81.0 % success rates on the first four models, while maintaining above 75.0 % effectiveness on the remaining three. This aggregated performance provides quantitative evidence of the IRAW method's generalization across diverse deep learning models.

4.2.3. Crafting invisible watermarks through diverse LOGO types and scales

A comprehensive visual comparison between the generated adversarial examples (G_1, G_2, G_3) and their host images (H_1, H_2, H_3) demonstrates negligible visual differences. This indicates that the proposed IRAW method maintains high-fidelity image quality regardless of the watermark type or scale. The effectiveness of the proposed method is substantiated by the altered class labels in the generated examples (G_1-G_3) compared to their host images (H_1-H_3), which successfully mislead well-trained DNNs into producing incorrect classifications. Fig. 5 (fourth column) shows text watermarks (THU/LOGO3, PKU/LOGO4)

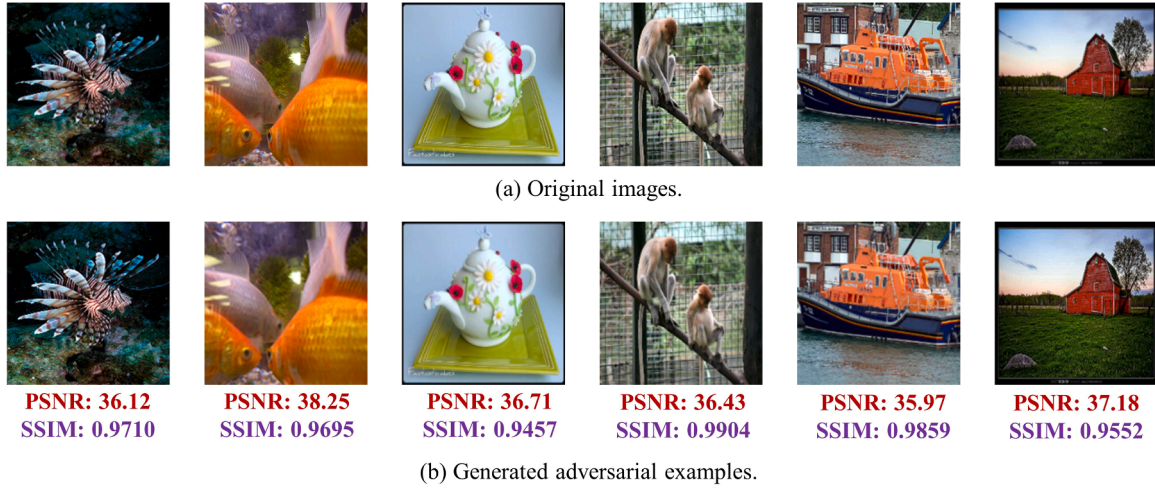


Fig. 4. Original images and adversarial examples generated by the IRAW method.

Table 2

Cumulative success rates of the IRAW method using LOGO1 and LOGO2 across six progressive stages.

Models		ShuffleNet	Inception_V3	ResNet50	VGG16	VGG19	ResNet101	ResNet152	Average
LOGO1 (%)	Stage1	52	56	59	65	67	54	61	59.1
	Stage2	68	64	63	72	70	59	63	65.6
	Stage3	71	69	75	77	72	67	65	70.9
	Stage4	81	76	78	79	74	70	72	75.7
	Stage5	87	82	80	84	76	74	72	79.3
	Stage6	94	84	82	85	78	76	74	81.9
LOGO2 (%)	Stage1	55	53	65	60	63	50	56	57.4
	Stage2	58	60	67	67	72	56	59	62.7
	Stage3	65	75	71	74	72	61	67	69.3
	Stage4	74	78	73	84	74	69	69	74.4
	Stage5	84	78	76	86	78	76	74	78.9
	Stage6	90	82	80	88	80	80	76	82.3
Average Success Rate of LOGO1 and LOGO2		92	83	81	86.5	79	78	75	82.1

embedded at 32×32 and 64×64 resolutions, all perfectly recovered by our extraction algorithm for copyright protection. This performance consistency extends to geometric watermarks, as demonstrated by LOGO5 implementations at varying scales. The shape watermark preserves its distinctive visual features after extraction, regardless of embedding dimensions. Results show that combining textual (LOGO3/LOGO4) and geometric (LOGO5) watermarks at varying scales achieves both adversarial perturbations and copyright protection without visual quality loss. These findings conclusively establish that our method produces adversarial examples whose visual fidelity remains uncompromised by watermark characteristics (type or size), while successfully fulfilling both primary objectives of adversarial effectiveness and content protection. The framework's robustness is further evidenced by consistent performance across all tested configurations, demonstrating its practical applicability in real-world UGC scenarios requiring secure yet imperceptible image modifications.

4.3. Comparisons with other attack methods

4.3.1. Visual quality of adversarial examples

Fig. 6 presents a comparative visualization of original images and their corresponding adversarial examples generated using various black-box perturbation approaches. Visual analysis reveals that Jia et al. (Jia, Wei, Cao, & Han, 2020) generates adversarial examples with conspicuous watermarks that are easily noticeable by human observers. Similarly, the spatial attack method, as well as the BHI (Liang, Feng, Chen, & Liu, 2023) and ISWP (Liang, Liu, Gao, Zhang, & Liu, 2025) techniques, produce perceptible alterations that are distinguishable from the original images. Furthermore, the pointwise attack method introduces

visually identifiable pixel-level modifications, which are clearly demarcated by red bounding boxes in the visualization (Fig. 6). In contrast, the proposed IRAW method generates adversarial examples with minimal perceptible differences from the source images, thereby demonstrating superior stealth and security performance. The average PSNR and SSIM metrics across the entire ImageNet dataset, as reported in Table 3, further confirm that both the pointwise attack method (42.03 dB / 0.9948) and our proposed IRAW method (33.21 dB / 0.9132) significantly outperform other competing schemes in preserving visual fidelity.

4.3.2. Attack performance of adversarial examples

A comprehensive evaluation of success rates is conducted on five prominent image classification models: Inception_V3 (Szegedy et al., 2016), VGG16 (Simonyan & Zisserman, 2014), ResNet50 (He, Zhang, Ren, & Sun, 2016), ResNet101 (He, Zhang, Ren, & Sun, 2016), and ResNet152 (He, Zhang, Ren, & Sun, 2016). The adversarial examples generated by various perturbation methods are systematically analyzed, with average success rates computed and presented in the final column of Table 5. These quantitative metrics enable effective comparison of performance across different black-box approaches. The single-pixel and pointwise approaches demonstrate limited efficacy, with success rates not exceeding 14 %. More robust performance is observed in spatial and boundary attack implementations. Notably superior results are attained using Jia et al.'s method (79.4 %) and ISWP (80.2 %), establishing them as particularly effective approaches. As quantitatively demonstrated in Table 4, IRAW method achieves superior performance over all comparable techniques, attaining a remarkable average success rate of 80.8 %. This empirical validation substantiates IRAW's outstanding

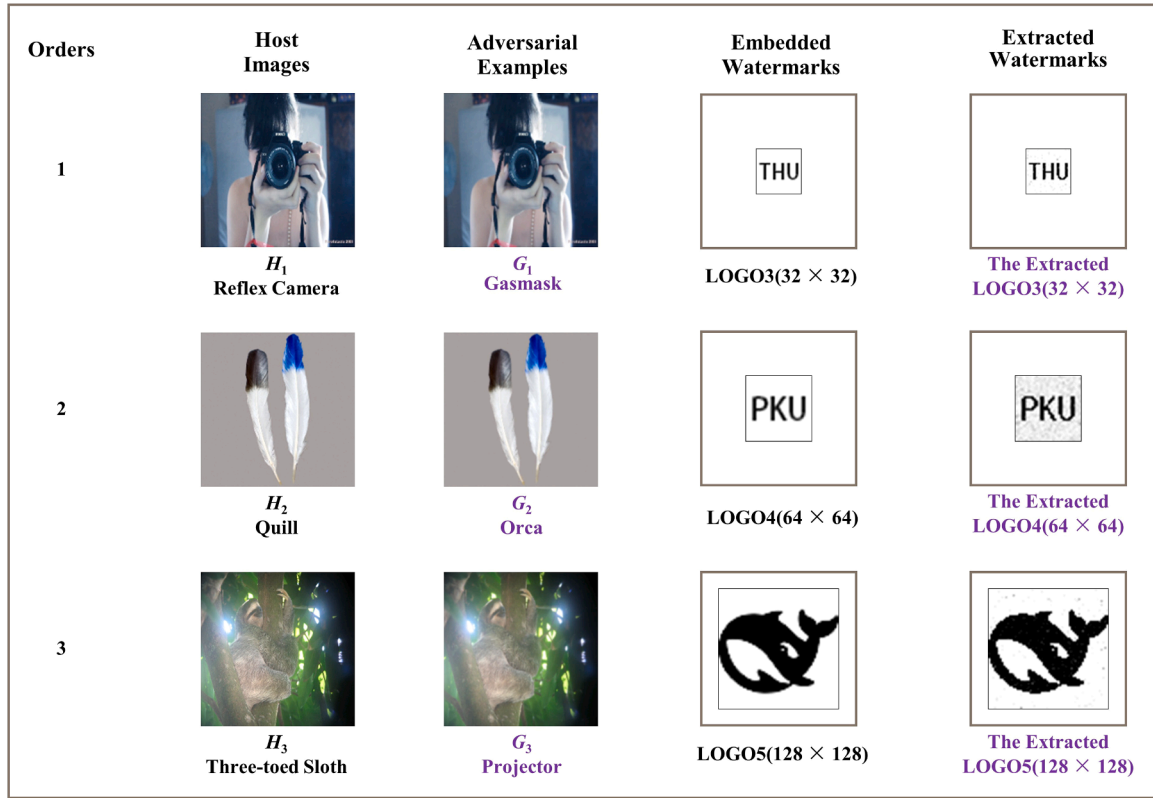


Fig. 5. The host images with their corresponding class labels and the embedded watermark information, represented in black text, are depicted below. The adversarial examples, including their predicted class labels, extracted watermarks, and associated size parameters, are all indicated using purple text annotations.

performance to generate adversarial examples that consistently evade detection by classification models, demonstrating significant improvements over current state-of-the-art methods in terms of the perturbation capability. We have extended our method to the CIFAR-10 dataset. This dataset is diverse and complementary to ImageNet, aiming to comprehensively evaluate the effectiveness and generalizability of the method. As quantitatively demonstrated in Table 5, the IRAW method outperforms all comparative approaches, achieving a notable average success rate of 81.2 %.

4.3.3. Robustness performance of adversarial examples

This section demonstrates a comprehensive comparative analysis between the proposed IRAW method and five established perturbation approaches: single-pixel attack (Engstrom, Tran, Tsipras, Schmidt, & Madry, 2017), boundary attack (Schott, Rauber, Bethge, & Brendel, 2018), Jia et al.'s method (Jia, Wei, Cao, & Han, 2020), BHI (Liang, Feng, Chen, & Liu, 2023), and ISWP (Liang, Liu, Gao, Zhang, & Liu, 2025). The evaluation incorporates ten standard image transformation operations (JPEG compression (Das et al., 2017), fragment, cloud, box blur, shape blur, Gaussian noise, Salt & Pepper, lens flare, crystallization, and resizing) to rigorously assess adversarial example robustness. Additionally, we have added more intensive JPEG compression tests (with quality factor (QF) set to 95 and 90, where a lower QF value indicates a higher compression intensity) as well as resizing tests with different dimensions (256×256 , 257×257). Robustness in this evaluation is defined as the capability of generated examples to maintain their adversarial capability after undergoing common image processing operations. To ensure statistical rigor, all success rates are calculated using all tested images from the ImageNet dataset as the constant denominator. Specifically, for a given perturbation method, the chosen transformation is applied to every generated adversarial sample; the resulting success rate represents the proportion of examples that remain adversarially effective relative to the total number of original test

images. For instance, the 79.4 % reported for BHI under salt-and-pepper noise (0.1) signifies that 79.4 % of all examples produced by BHI successfully maintained their adversarial effectiveness after the entire dataset was subjected to the noise.

Experimental results reveal significant performance variations among the evaluated methods, as shown in Table 6. The results indicate that our proposed IRAW (DCT) method and ISWP (DWT) generally outperform Jia et al.'s method and BHI, as transform-domain perturbations are inherently more resilient to common filters and distortions than spatial patterns. By embedding signals into the mid-frequency coefficients, our method achieves a stable coupling with the image's structural features, allowing the perturbation information to survive even under severe environmental noise. In contrast, the robustness of boundary and single-pixel approaches is significantly lower, as they rely on fragile decision boundaries and precise pixel values that are easily altered during transformation operations. Furthermore, the experimental results reveal that adversarial robustness is highly dependent on the alignment between the perturbation embedding domain and the model's feature extraction preference. For Inception_V3, the frequency-domain invisible watermarks exhibit superior resilience because they manipulate global spectral features that coincide with the multi-scale kernels of Inception modules. Conversely, in ResNet101, the residual structures are more sensitive to localized semantic patterns, leading to the relatively competitive performance of visible watermarks. This confirms that the proposed transform-domain approach provides a more robust and adaptive black-box method across varying neural architectures.

4.3.4. Extended robustness evaluation of invisible watermarks in IRAW

To quantitatively evaluate the extraction quality of invisible watermarks, we employ normalized correlation (NC) as the primary metric. Specifically, a binary geometric watermark (with 32×32 resolution) is embedded into the host image to generate an adversarial example that is



Fig. 6. Comparative visualization of original images and their corresponding adversarial examples generated by various black-box perturbation methods.

Table 3

Comparison of average visual quality across the entire ImageNet dataset for various black-box perturbation methods.

Methods	Pointwise Attack	Jia et al.'s Method	Spatial Attack	BHI	ISWP	The Proposed Method
PSNR (dB)	42.03	20.59	12.75	30.37	31.04	33.21
SSIM	0.9948	0.8026	0.2429	0.9017	0.8989	0.9132

then subjected to the fourteen standard image transformation operations detailed in Table 6. Following these attacks, the watermarks are extracted, and their fidelity is assessed by calculating the NC between the extracted and the original watermark. The results summarized in Table 7 demonstrate that our method achieves high fidelity in watermark recovery, with an average NC exceeding 0.9261. Even when adversarial samples are subjected to severe distortions such as JPEG

compression, resizing, and Gaussian noise, the extracted watermarks remain highly identifiable.

Three significant phenomena can be observed in Table 7: First, Inception_V3 achieves a superior average NC of 0.9333 compared to ResNet101 (0.9261). This suggests that the multi-scale convolutional kernels inherent in the inception module possess higher sensitivity and precision in capturing frequency-domain features, which not only facilitates higher adversarial transferability (as seen in Table 6) but also ensures more complete recovery of the watermark for copyright protection under common signal processing interferences. Second, the NC values remain consistently high against these image attack approaches, especially in cases of JPEG compression or blurring. This is not due to improved visual quality, but because these operations act as non-linear low-pass filters that remove high-frequency noise and redundant textures, mitigating the masking effect of the host image and optimizing the detection signal-to-noise ratio (SNR). Third, resizing attacks reveal a distinct performance gap between even and odd dimensions; even dimensions (e.g., 226×226) yield significantly higher NC values than odd ones (e.g., 225 or 227) due to superior pixel alignment symmetry during

Table 4

The attack rates of adversarial examples generated by different black-box perturbation approaches on ImageNet.

Models	Inception_V3 (%)	VGG16 (%)	ResNet50 (%)	ResNet101 (%)	ResNet152 (%)	Average (%)
Single-pixel Attack	5	13	9	5	8	8
Pointwise Attack	7	21	21	7	14	14
Boundary Attack	48	48	8	37	6	29.4
Spatial Attack	58	84	78	52	71	68.6
BHI	81	79	67	78	54	71.8
Jia et al.'s Method	73	83	83	78	80	79.4
ISWP	81	85	83	75	77	80.2
The Proposed Method	83	87	81	78	75	80.8

Table 5

The attack rates of adversarial examples generated by different black-box perturbation approaches on CIFAR-10.

Models	VGG16 (%)	ResNet18 (%)	ResNet50 (%)	ResNet101 (%)	ResNet152 (%)	Average (%)
<i>Single-pixel Attack</i>	18	16	13	11	12	14
<i>Pointwise Attack</i>	38	22	19	15	17	22.2
<i>Boundary Attack</i>	55	43	31	26	22	35.4
<i>Spatial Attack</i>	76	68	60	56	48	61.6
<i>BHI</i>	84	78	71	77	69	75.8
<i>Jia et al.'s Method</i>	87	73	76	79	81	79.2
<i>ISWP</i>	83	81	84	76	78	80.4
<i>The Proposed Method</i>	87	78	84	79	78	81.2

Table 6

The robustness of adversarial examples generated by different black-box perturbations methods.

Methods (%)		Boundary Attack (%)	Single-pixel Attack (%)	Jia et al.'s Method (%)	BHI (%)	ISWP (%)	The Proposed Method (%)
Inception_V3	JPEG compression (QF = 100)	3.8	2.1	68.6	68.0	79.4	83.0
	JPEG compression (QF = 95)	3.4	1.8	66.4	64.8	77.0	81.3
	JPEG compression (QF = 90)	1.9	1.6	65.0	64.8	72.9	78.9
	Fragment (1)	33.1	2.0	70.1	79.4	80.2	82.2
	Cloud (1)	33.1	1.0	69.4	79.4	80.2	80.2
	Box blur (2)	37.4	4.0	70.1	73.7	80.2	82.2
	Shape Blur (5)	22.1	1.0	70.1	73.7	79.4	81.3
	Gaussian noise (0.1)	19.7	2.8	69.4	79.4	81.0	81.0
	Salt & Pepper (0.1)	14.9	4.0	69.4	79.4	81.0	82.2
	Lens flare (3)	25.9	2.0	70.1	79.4	80.2	80.2
	Crystallization (9)	18.2	4.0	70.1	75.3	80.2	82.2
	Resizing (225 × 225)	9.6	0.4	66.4	67.2	77.0	78.0
	Resizing (226 × 226)	13.4	0.8	67.9	68.9	78.9	78.6
	Resizing (227 × 227)	20.2	1.6	69.4	71.3	79.4	80.5
	Average	18.3	2.1	68.7	73.2	79.1	80.8
ResNet101	JPEG compression (QF = 100)	4.8	1.2	72.5	68.6	77.2	77.2
	JPEG compression (QF = 95)	4.1	1.0	70.2	66.3	72.5	74.9
	JPEG compression (QF = 90)	3.0	0.8	68.6	63.2	71.0	71.0
	Fragment (1)	18.5	4.3	74.9	70.2	72.8	77.2
	Cloud (1)	27.8	5.0	75.7	72.5	72.8	75.7
	Box blur (2)	4.8	2.2	74.9	64.7	74.3	77.2
	Shape blur (5)	14.1	3.6	75.7	64.7	75.7	75.7
	Gaussian noise (0.1)	22.2	4.3	75.7	72.5	75.7	75.7
	Salt & Pepper (0.1)	14.1	4.3	75.7	68.6	72.8	75.7
	Lens flare (3)	27.8	4.3	72.5	70.2	72.5	72.5
	Crystallization (9)	9.3	2.9	74.9	63.2	72.8	76.4
	Resizing (225 × 225)	15.9	0.7	67.1	40.6	67.1	66.3
	Resizing (226 × 226)	14.1	0.5	66.3	37.4	63.2	63.2
	Resizing (227 × 227)	10.7	0.3	63.2	33.5	62.3	62.3
	Average	13.7	2.5	72.0	61.2	71.7	72.9

Table 7

Normalized correlation (NC) values between extracted and original watermarks under typical image attacks.

Models	ResNet101	Inception_V3
JPEG compression (QF = 100)	0.9292	0.9370
JPEG compression (QF = 95)	0.9292	0.9370
JPEG compression (QF = 90)	0.9215	0.9296
Fragment (1)	0.9213	0.9292
Cloud (1)	0.9213	0.9261
Box blur (2)	0.9292	0.9370
Shape blur (5)	0.9235	0.9339
Gaussian noise (0.1)	0.9215	0.9292
Salt & Pepper (0.1)	0.9213	0.9356
Lens flare (3)	0.9215	0.9296
Crystallization (9)	0.9139	0.9220
Resizing (225 × 225)	0.9215	0.9296
Resizing (226 × 226)	0.9607	0.9528
Resizing (227 × 227)	0.9296	0.9372
Average	0.9261	0.9333

spatial interpolation, which minimizes phase shifts and energy dispersion. These quantitative findings, combined with the sustained adversarial effectiveness discussed previously, provide empirical evidence for the robustness of our dual-protection framework. This confirms that the

proposed method can simultaneously maintain defensive potency against AI-driven exploitation while ensuring reliable ownership traceability under complex image transformations.

4.3.5. Ablation studies

We have conducted targeted ablation experiments to quantify the contributions of each core component. Specifically, we constructed three sets of comparative schemes: a basic scheme adopting only the basin hopping (BH) algorithm, a combined scheme integrating BH and discrete cosine transform (DCT) coefficient optimization, and the complete IRAW scheme incorporating BH, DCT coefficient selection, and cross-domain encryption. By evaluating the success rates of the adversarial perturbations of the above three schemes on target classification models including Inception-V3, VGG16, ResNet50, ResNet101, and ResNet152 using the original ImageNet test set, we formulated a dedicated ablation experiment comparison table (Table 8). The ablation results in Table 8 show that the scheme adopting only the BH algorithm achieves an average success rate of 58.6 %, the scheme integrating BH and DCT coefficient optimization increases the average success rate to 71.4 %, and the complete IRAW scheme incorporating BH, DCT, and cross-domain encryption further improves the average success rate to 80.8 %. The inclusion of the cross-domain encryption module not only fulfills the security requirement of preventing unauthorized access but

Table 8

Ablation studies on key components of the proposed IRAW method.

Models	Inception_v3(%)	VGG16 (%)	ResNet50 (%)	ResNet101 (%)	ResNet152 (%)	Average (%)
BH Only (%)	55	67	63	58	50	58.6
BH + DCT Optimization (%)	76	81	74	65	61	71.4
BH + DCT + Cross-Domain Encryption (Complete IRAW) (%)	83	87	81	78	75	80.8

also contributes to the enhanced perturbation performance. This phenomenon can be attributed to the non-linear mappings within the DCT domain, which alter the statistical distribution of the perturbation signals and increase the complexity of the adversarial patterns. Furthermore, the stochasticity introduced by the encryption process assists the BH algorithm in exploring a broader search space, thereby facilitating the escape from local optima and converging toward a more effective global solution. The gradual integration of each core module has achieved a significant improvement in perturbation performance, fully verifying the rationality and necessity of the combined design.

4.4. Results analysis and discussion

The proposed IRAW method fundamentally transforms conventional adversarial methods through generating adversarial examples with invisible watermarks. The superior characteristics of this method are listed as follows:

- **Dual-Functional Fusion:** IRAW addresses a dual-functional challenge by concurrently executing adversarial perturbations and implementing copyright protection through invisible watermarks.
- **High Success Rate:** As shown in Table 4 and Table 5, the IRAW method achieves significantly higher success rates than conventional black-box perturbation approaches, including spatial attack, boundary attack, single-pixel attack, pointwise attack, Jia et al.'s method, BHI, and ISWP.
- **Good Visual Performance:** The adversarial examples embedded with imperceptible watermark perturbations maintain excellent visual fidelity, showing negligible perceptual differences from their original host images that are virtually undetectable to human observers (Fig. 1(b) and Fig. 4).
- **Robustness:** As evidenced in Table 6, these adversarial examples generated by IRAW demonstrate stronger resilience against diverse image transformation operations compared with state-of-the-art adversarial methods.
- **Security:** IRAW implements a cross-domain encryption system featuring dual spatial-domain encryption for both host images and watermarks, combined with frequency-domain coordinate obfuscation. The generated adversarial examples retain visual appearance to evade human detection, yet incorporate multiple security safeguards. To enhance security against adaptive attacks, the extraction keys and parameters are pre-encrypted using a public key before embedding. Under this threat model, even if the specific embedding locations are identified, attackers cannot obtain the actual data without the authorized private key. This ensures the system's security is anchored in cryptographic strength rather than location secrecy.
- **Meaningful Perturbations:** Compared to conventional adversarial methods that generate arbitrary noise patterns, the perturbations produced by IRAW carry semantically meaningful information, enabling the simultaneous execution of adversarial perturbations and copyright protection.

5. Conclusion

In this paper, we propose IRAW, an innovative proactive image protection mechanism designed to transition digital asset security from

passive traceability to active defense. By integrating invisible watermarking with adversarial perturbations within the discrete cosine transform (DCT) domain, IRAW ensures native compatibility with the JPEG compression protocols essential for mainstream social media and user-generated content (UGC) platforms. The proposed method effectively neutralizes unauthorized AI-driven analysis by preventing illicit scrapers from automatically categorizing or indexing private media, thereby mitigating large-scale data exploitation. To enhance the security of IRAW, we employ a dual encryption strategy in the spatial domain for both watermark and host image data, complemented by frequency domain coordinate obfuscation. This integrated methodology establishes a cross-domain protection architecture, actively defending copyrighted content in both spatial and spectral dimensions. Furthermore, the introduction of this cross-domain mechanism not only bolsters system security but also increases the complexity of the perturbations, thereby enabling BH to effectively escape local optima and significantly enhance the perturbation effectiveness of the generated adversarial examples. Experimental results demonstrate that the generated adversarial examples present benign visual quality with relatively robust security performance. The perturbations in adversarial examples are visually indistinguishable from the original images to human observers, which provides an effective safeguard against potential image operations. Compared with widely adopted adversarial approaches, the adversarial perturbations generated by the proposed IRAW achieve superior success rates and robustness. Remarkably, IRAW successfully fulfills the dual objectives of both adversarial perturbations and copyright protection. As an effective and industrially compatible solution, IRAW is poised to address emerging privacy and security risks in UGC management. In the future, we will redesign the algorithm to generate adversarial examples with invisible watermarks that are more robust and visually superior, thereby protecting digital assets in the metaverse.

CRedit authorship contribution statement

Jinchao Liang: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Yangcheng Chen:** Writing – original draft, Software, Formal analysis, Data curation. **Shuwu Chen:** Writing – review & editing, Validation, Project administration, Formal analysis. **Xiaolong Liu:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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